Dr Vivek Nangia, principal director and head of pulmonology at the Max Super Speciality Hospital in Saket, pollutants now "can trigger an inflammatory cascade that can impact the brain, kidneys, heart, and even cause diabetes". The allergy-like symptoms are only the immediate and short-term consequence of the poor air quality index (AQI). The longer-term impact is more insidious because it is invisible.

It is costing the average Indian a few years of his or her life. According to this year's Global Air Quality Life Index report released by the Energy Policy Institute at the University of Chicago, the current levels of pollution, particularly fine particulate matter, will stave 5.3 years off from the average Indian's life. Delhizens will lose some more, since living in the most polluted city in the country will cost them 11.9 years. Compared to this, cardiovascular diseases reduce the average Indian's life expectancy by about 4.5 years, child and maternal malnutrition by 1.8 years and tobacco use by 2.2 years. When the then US president Barack Obama came to India for three days to attend the Republic Day parade in 2015, on his return, David Spiegelhalter, a statistician at the University of Cambridge who specialises in quantifying risk, calculated that his life span may reduce by six hours since you lose roughly two hours of your life a day when you breathe this kind of air.

"All living things need good quality air to survive," says Dr Arjun Khanna, head of the department of pulmonary medicine at Amrita Hospitals, Faridabad. "When this is polluted, life itself is threatened." So, what do you do about it? Understand the nature of the beast first. Which means you ask...

WHAT IS POLLUTION?

Pollution is what you get when toxic materials enter the environment. Today's pollution is usually a mixture of highly concentrated carbon monoxide, nitrogen oxide, sulfur dioxide and

WHAT IS POLLUTION?

Anything toxic in air, water, or any substance is a pollutant. Air pollution is caused when cars, factories and the burning of fossil fuels produce highly concentrated carbon monoxide, nitrogen oxide, sulphur dioxide and hydrocarbons. These react with sunlight to produce smog, a thick fog or haze

WHAT IS PARTICULATE MATTER?

Not always visible, it comprises particles of chemicals, soil, smoke, dust or allergens. Different in size, shape and chemical composition, these particles are defined by their diameter for air quality regulatory purposes. The tiniest of these particles (PM2.5) are 30 times thinner than a strand of human hair and can easily enter our bodies through our airways

POLLUTION AND ITS IMPACT ON VARIOUS ORGANS

LUNGS

Particulate matter enters our airways first and causes an immune reaction we recognise as inflammation. This ongoing inflammation is a leading risk factor for COPD and lung cancer

HEART

Particulate matter can prematurely age our blood vessels and lead to more build-up of calcium in the arteries—a risk for stroke and heart attack. It can also cause changes to the heart rhythm and structure

HOW TO STAY SAFE



Wear a well-fitted N-95 mask every time you step outdoors

Avoid outdoor exercise, crowded spaces and malls



Try to stay indoors as much as possible. Air filters might help but only if the room is significantly sealed of outside air

Graphics by TANMOY CHAKRABORTY